

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTECOE504C - PROGRAMMING IN JAVA

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Java is a:
 - a) Compiled language only
 - b) Interpreted language only
 - c) Both compiled and interpreted
 - d) Assembled languageWhich of the following is used to create an object in Java?
 - a) create
 - b) new
 - c) alloc
 - d) construct
- 3 What is the purpose of the 'this' keyword in Java?
 - a) Refers to the superclass
 - b) Refers to the current class
 - c) Refers to the current object
 - d) Refers to the child class
- 4 Which method is called automatically when an object is created?
 - a) finalize()
 - b) main()
 - c) init()
 - d) Constructor
- 5 What is the use of the 'implements' keyword?
 - a) To inherit from a class
 - b) To define a package
 - c) To implement an interface
 - d) To import a class
- 6 Can a class implement multiple interfaces in Java?
 - a) No
 - b) Only if interfaces are final
 - c) Yes
 - d) Only one interface at a time

7 Which interface must be implemented to create a thread in Java?

- a) Runnable
- b) Threadable
- c) Callable
- d) EventListener

8 Which keyword is used to handle exceptions in Java?

- a) try-catch
- b) exception
- c) catch-throw
- d) error-handle

Which package is primarily used for graphics in Java?

- a) java.awt
- b) java.io
- c) java.util
- d) java.graphics

Which method is used to load a JDBC driver class in Java?

- a) DriverManager.connect()
- b) Class.forName()
- c) loadDriver()
- d) Driver.load()

11 Which one of the following is not an access modifier?

- a) Protected
- b) Void
- c) Public
- d) Private

What is the output of the following code?

```
class A {  
    int x = 10;  
}  
  
public class Test {  
    public static void main(String[] args) {  
        A obj = new A();  
        System.out.println(obj.x);  
    }  
}
```

- a) x
- b) 0
- c) 10
- d) Compile error

Q2. Solve the following.

Differentiate between C++ and Java.

6

What is variable scope in Java? Discuss local, instance, and static variables with suitable examples.

6

Q3. Solve the following.

- A) Explain the concept of 'this' keyword in Java. How is it used to refer to current class members and constructor chaining? 6
- B) Explain the use of the `final` keyword in Java with respect to variables, methods, and classes. What restrictions does it impose in each case? 6

Q4. Solve Any Two of the following.

- A) State the difference between inheritance and interface in java. 6
- B) Explain the concept of method overriding in Java. Illustrate with an example. 6

- C) What is inheritance in Java? List the types of inheritance supported in Java and explain multilevel inheritance with a suitable example. 6
- Q5. Solve Any Two of the following.
- A) What is multithreading in Java? Explain the lifecycle of a thread with a suitable diagram. 6
- B) What is exception handling in Java? Explain the use of try, catch, and finally, keywords with an example. 6
- C) Differentiate between checked and unchecked exceptions in Java. 6
- Q6. Solve Any Two of the following.
- A) What is JDBC in Java? Explain the steps to connect a Java application to a database using JDBC with example. 6
- B) What is the Abstract Window Toolkit (AWT) in Java? Explain the key features of the AWT package. 6
- C) Explain the Graphics class in Java. How is it used to draw shapes and text in GUI applications? 6

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